Supplementary Material for “Not All Embeddings Are Gaussian: Revisiting Adaptive HNSW Search”

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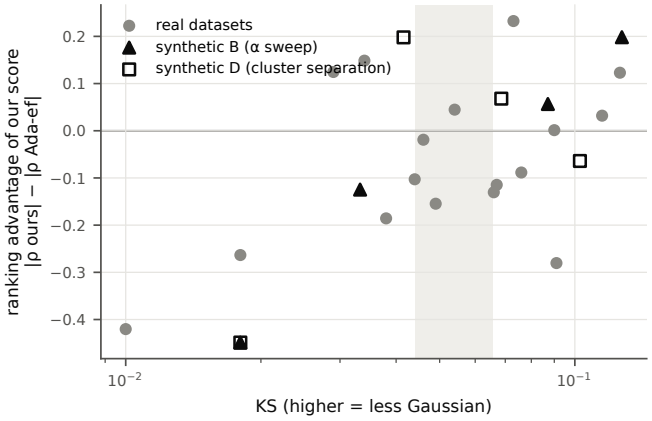


Fig. S1. Ranking advantage of the empirical score (its $|\rho|$ minus Ada-ef's) against KS, for the real datasets and for synthetic data with controlled anisotropy (B) and cluster separation (D).

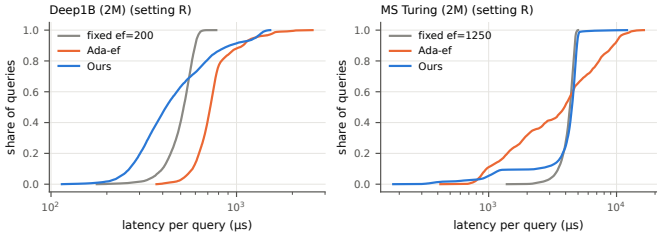


Fig. S2. Per-query latency on Deep1B and MS Turing (setting R). The empirical score spends less time on easy queries and more on hard ones; a fixed *ef* spends about the same on all.

TABLE S1

Time per distance computation, relative to fixed-*ef* queries with the same distance count (settings P and R). Hard and easy: the 20% of queries PercEF gives the largest and smallest *ef*, timed under the fixed *ef* closest to PercEF's mean recall. Last column: PercEF's share of its distance computations spent on its hard queries.

dataset	setting	KS	hard / fixed-ef time at same DC	easy / fixed-ef time at same DC	PercEF / fixed-ef time at same DC	PercEF DC on hard queries (%)
MS Turing (2M)	P	0.010	1.009	1.003	0.991	95.1
MS Turing (2M)	R	0.010	1.003	0.997	0.981	97.9
GloVe-100	P	0.018	1.036	0.803	1.025	95.7
GloVe-100	R	0.018	1.036	0.803	1.014	97.3
BIGANN (2M)	P	0.029	1.076	0.807	1.032	50.5
BIGANN (2M)	R	0.029	1.087	0.800	1.025	39.0
LAION-I2I	P	0.034	1.191	0.842	1.043	58.0
LAION-I2I	R	0.034	1.174	0.809	1.057	54.3
DBpedia-1536	P	0.038	1.048	0.885	1.000	81.2
DBpedia-1536	R	0.038	1.057	0.834	1.036	66.1
MS MARCO-384	P	0.044	1.132	0.816	1.004	30.4
MS MARCO-384	R	0.044	1.134	0.827	0.971	30.8
COCO-I2I	P	0.046	1.148	0.925	1.053	43.8
COCO-I2I	R	0.046	1.168	0.908	1.051	35.9
Cohere-1024	P	0.049	1.049	0.801	0.976	63.7
Cohere-1024	R	0.049	1.066	0.837	1.027	71.2
Yahoo-MiniLM	P	0.054	1.084	0.890	1.022	44.9
Yahoo-MiniLM	R	0.054	1.088	0.893	1.029	39.8
COCO-T2I	P	0.056	1.005	1.005	1.011	100.0
COCO-T2I	R	0.056	1.001	1.001	1.014	100.0
ImageNet-ALIGN	P	0.058	1.009	1.009	1.031	100.0
ImageNet-ALIGN	R	0.058	1.000	1.000	1.027	100.0
DeepImage-96	P	0.066	1.074	0.916	1.066	52.0
DeepImage-96	R	0.066	1.089	0.910	1.076	43.6
Deep1B (2M)	P	0.067	1.092	0.889	1.053	34.4
Deep1B (2M)	R	0.067	1.084	0.882	1.074	43.9
Fashion-MNIST	P	0.073	1.093	0.980	1.017	30.3
Fashion-MNIST	R	0.073	1.091	0.978	1.015	25.6
Landmark-DINO	P	0.076	1.085	0.893	0.976	52.9
Landmark-DINO	R	0.076	1.090	0.892	0.997	49.0
Yambda	P	0.090	1.004	1.002	0.977	100.0
Yambda	R	0.090	0.999	0.978	0.989	100.0
GIST-960	P	0.091	1.005	1.002	0.977	98.8
GIST-960	R	0.091	1.005	1.002	0.981	99.4
iNat-ResNet	P	0.115	1.109	0.930	0.993	33.7
iNat-ResNet	R	0.115	1.123	0.912	1.000	30.0
SIFT-1M	P	0.126	1.050	0.896	1.005	42.7
SIFT-1M	R	0.126	1.060	0.894	1.006	28.9
Last.fm	P	0.216	0.999	0.999	0.985	100.0
Last.fm	R	0.216	0.996	0.996	0.994	100.0

TABLE S2
Offline cost.

dataset	setting	Ada-ef statistics (s)	Ada-ef table (s)	Ada-ef memory (B)	ours bins (s)	ours min-ef (s)	ours memory (B)
MS Turing (2M)	P	5.2	5.2	41,368	0.6	7.4	824
MS Turing (2M)	R	5.2	183.3	41,504	0.6	257.5	768
GloVe-100	P	–	36.6	41,392	0.4	–	672
GloVe-100	R	–	370.4	41,608	0.4	–	820
BIGANN (2M)	P	4.8	0.3	67,240	1.3	0.3	936
BIGANN (2M)	R	4.8	3.0	67,344	1.3	3.3	936
LAION-I2I	P	–	21.9	1,053,216	–	–	2,472
LAION-I2I	R	–	347.9	1,053,480	–	–	2,472
DBpedia-1536	P	–	1.0	9,450,000	–	–	6,540
DBpedia-1536	R	–	14.1	9,450,176	–	–	6,540
MS MARCO-384	P	–	4.5	593,160	–	–	1,952
MS MARCO-384	R	–	45.4	593,432	–	–	1,960
COCO-I2I	P	1.7	0.1	1,053,344	0.2	0.1	2,472
COCO-I2I	R	1.7	1.0	1,053,448	0.2	1.1	2,472
Cohere-1024	P	–	6.0	4,202,768	–	–	4,520
Cohere-1024	R	–	69.9	4,203,000	–	–	4,476
Yahoo-MiniLM	P	–	0.3	593,328	–	–	1,912
Yahoo-MiniLM	R	–	0.6	593,384	–	–	1,836
COCO-T2I	P	1.7	0.1	1,053,352	0.2	0.1	2,472
COCO-T2I	R	1.7	7.0	1,053,416	0.2	9.7	2,072
ImageNet-ALIGN	P	–	0.1	1,643,904	–	–	2,984
ImageNet-ALIGN	R	–	26.7	1,644,232	–	–	2,584
DeepImage-96	P	–	0.8	37,912	3.1	–	808
DeepImage-96	R	–	9.5	38,192	3.1	–	808
Deep1B (2M)	P	5.8	0.3	38,080	0.6	0.2	808
Deep1B (2M)	R	5.8	3.4	38,336	0.6	3.3	808
Fashion-MNIST	P	–	0.1	2,465,040	–	–	3,560
Fashion-MNIST	R	–	0.8	2,465,384	–	–	3,560
Landmark-DINO	P	–	0.6	2,365,928	–	–	3,496
Landmark-DINO	R	–	0.8	2,365,976	–	–	3,496
Yambda	P	–	0.1	66,608	–	–	936
Yambda	R	–	0.8	66,704	–	–	936
GIST-960	P	–	7.0	3,694,232	–	–	3,888
GIST-960	R	–	13.3	3,694,208	–	–	3,888
iNat-ResNet	P	–	0.4	16,794,184	–	–	8,616
iNat-ResNet	R	–	0.7	16,794,224	–	–	8,616
SIFT-1M	P	–	0.1	66,776	0.4	–	936
SIFT-1M	R	–	1.4	67,224	0.4	–	936
Last.fm	P	0.8	0.3	17,444	0.1	2.1	684
Last.fm	R	0.8	0.3	17,804	0.1	0.4	284